AI PROCTORING OF ONLINE EXAMINATIONS

Harare Institute of Technology
success through innovation

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1.0 Abstract

Abstract - Online education has necessitated effective exam proctoring to maintain integrity. This research introduces an AI Proctoring system leveraging computer vision techniques for real-time analysis of video feeds during online exams. The system's architecture, implementation, evaluation, and results are presented, demonstrating its ability to identify cheating behaviors accurately. Limitations and ethical considerations are discussed. These findings contribute to developing robust AI Proctoring systems for fair assessment in online education.

2.0. Introduction

Cheating detection involves analyzing eye movements, facial expressions, head movements, and typing behaviors. AI proctors observe and analyze these cues using algorithms and technology. Techniques like eye tracking, head movement analysis, and text input analysis aid decision-making. AI proctors overcome limitations of manual proctoring, monitoring all students simultaneously. Detecting suspicious actions helps identify potential academic dishonesty during online exams.

3.0. Problem definition

The rise of online education has highlighted the need for effective proctoring to maintain exam integrity and prevent cheating. At the Harare Institute of Technology (HIT), which uses a hybrid model of online and face-to-face classes, traditional in-person proctoring methods are insufficient, leading to increased cheating in online exams. Reliable and scalable proctoring solutions are essential for educational institutions to ensure fair and credible assessments. Manual proctoring is inefficient and prone to human error, underscoring the need for automated systems.

Privacy and security concerns must be balanced with effective monitoring to gain user trust and comply with regulations. The AI Proctoring System project aims to develop a scalable, automated solution using computer vision, machine learning, and audio analysis to monitor exams, detect cheating, and verify test-taker identity. The system will prioritize privacy, security, and ethical considerations, providing a reliable, efficient, and secure solution for online exam monitoring. This will enhance exam integrity, deter academic dishonesty, and maintain the value of educational certifications in the digital era.

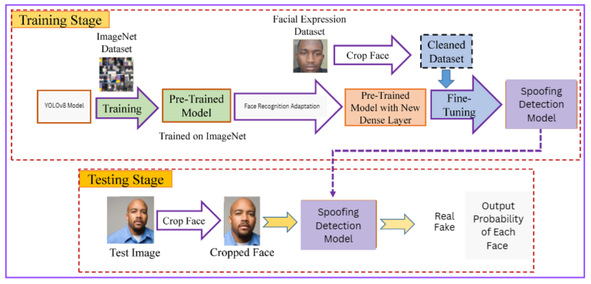
4.0. Overview of the developed application

The AI Proctoring system integrated into a Learning Management System (LMS) ensures seamless and automated monitoring of online exams to maintain academic integrity and prevent cheating. This system is embedded within the LMS, allowing easy access for both instructors and students, and automates the scheduling and initiation of proctored exams directly through the LMS interface. It uses facial recognition and biometric verification to authenticate the identity of test-takers before and during exams.
The system employs computer vision to analyze video feeds from students' webcams, monitors audio through microphones to detect unauthorized noises or conversations, and tracks screen activity and keystrokes to identify potential cheating behaviors. Suspicious activities such as the presence of unauthorized materials, multiple faces in the frame, or unusual patterns in eye movement and facial expressions are detected, and alerts are generated for accessing external websites, using unauthorized devices, or communicating with others during the exam. Machine learning algorithms analyze collected data to detect anomalies and provide detailed reports to instructors, including flagged incidents and summaries of activities. The system ensures compliance with privacy regulations by securely storing and encrypting data, providing transparency about data collection, usage, and retention policies, and implementing measures to protect user confidentiality

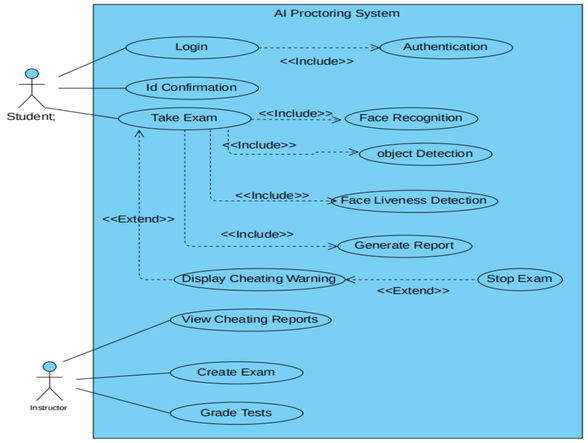
5.0. Objectives

The goal is to develop a method of determining the probability of cheating behaviors and their accuracy. Specifically, the goal is to detect "proxy test taker" behavior and identify specific indicators of this form of cheating with the following objectives:
To use machine learning techniques (facial recognition, spoofing detection, head pose estimation, cell phone detection) to detect various cheating actions during exam sessions.
To allow users to create and enrol for courses and exams online
To block certain website access during proctoring session
To alert examiner if any abnormal behavior is detected

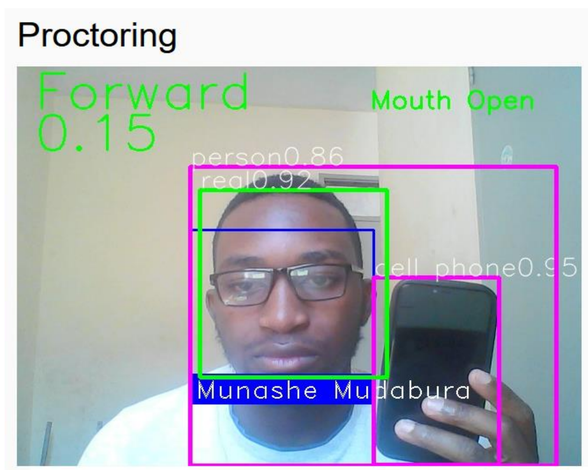
6.0. System Model Training

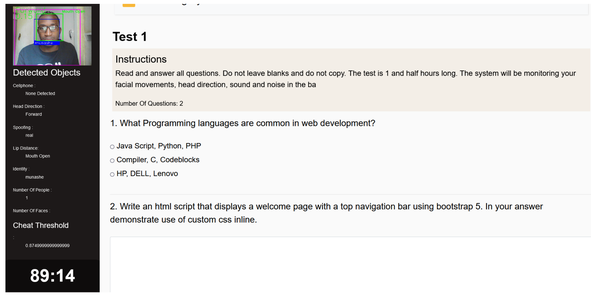


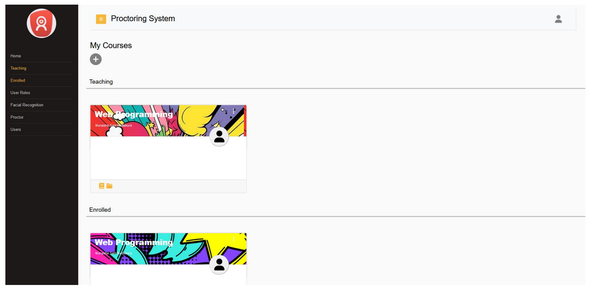
7.0. Use Case Diagram



8.0. Screenshots







Conclusion

The AI Proctoring System for online examinations has demonstrated significant potential in maintaining academic integrity through the use of advanced machine learning algorithms and computer vision techniques. The system has shown efficacy in monitoring and analyzing various aspects of student behavior, such as facial recognition, cellphone detection, multiple face detection, noise detection, and head pose estimation.

In recent years, researchers have developed deep learning models and neural networks that have greatly enhanced the accuracy and reliability of these proctoring systems. State-of-the-art models in areas such as facial recognition and object detection have achieved high levels of accuracy, providing robust solutions for detecting cheating during online exams.
However, there remains substantial room for improvement.

Enhancing the system's robustness in diverse environmental conditions, such as varying lighting, camera qualities, and background noise, is essential. Additionally, integrating new features like keyboard activity monitoring, browser lockdown, and eye-tracking can further improve the system's capability to detect and prevent cheating. Real-time detection and response also pose ongoing challenges that need to be addressed to ensure that alerts are immediate and accurate.

Despite these challenges, the AI Proctoring System has the potential to revolutionize the way online examinations are conducted, significantly enhancing academic integrity. The system can be applied across various fields of education, from primary to higher education, and in professional certification programs. Ongoing research and development are crucial for improving the system's performance and expanding its applicability. This includes exploring new AI techniques, refining existing algorithms, and addressing ethical and privacy concerns to ensure that the system is both effective and respectful of users' rights.

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